



Facilitator Guide

Sector
Green Jobs

Sub Sector
Renewable Energy
Occupation
Entrepreneur

Reference Id
SGJ/Q0901, Version 2.0; NSQF Level 5



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Solar PV Entrepreneur

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“

Skilling is building a better India.
If we have to move India towards
development then Skill Development
should be our mission.

”

Shri Narendra Modi
Prime Minister of India





Certificate

COMPLIANCE TO QUALIFICATION PACK – NATIONAL OCCUPATIONAL STANDARDS

is hereby issued by the

SKILL COUNCIL FOR GREEN JOBS

for the

SKILLING CONTENT: FACILITATOR GUIDE

Complying to National Occupational Standards of

Job Role/ Qualification Pack: "Solar Photovoltaic Entrepreneur" QP No. 'SGJ/ Q 0901
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Acknowledgement



**Institute of Solar Power
Technologies & Vocational
Training Development**



Kontent Edge Private Limited

About this book

Government of India is aiming towards a capacity installation of over 300 GW of solar PV out (includes both rooftop and ground mounted projects) of cumulative 450 GW of renewable energy targets by 2030. In 2015, a target of 175 GW of renewable energy capacity was planned to be achieved by 2022 out of which 100 GW was from Solar Energy (40 GW by Solar rooftop and 60 GW by Ground mounted projects). As per an estimate of a recent study carried out by CEEW-NRDC and SCGJ, over 3 Million new jobs can be created in domestic renewable energy sector (mainly solar and wind energy) by 2030. Considering the huge technically trained workforce requirement to meet this ambitious goal, Skill Council for Green Jobs has designed and implemented a range of skilling interventions with a focus on Project construction, system Installation and Operations & Maintenance related job roles for both solar rooftop and ground mounted solar projects. Solar PV Entrepreneur is a key job role who ventures into Solar market to lead an enterprise. He/She has the understanding of solar business models, market, technical knowledge of rooftop solar PV plants, along with components procurement and financing. Solar Photovoltaic Entrepreneur can prepare the feasibility study report and perform basic energy generation forecasting using simulation software and is also responsible for the managing the complete Solar PV project lifecycle. He/she is required by the solar industry to drive capacity addition in both solar rooftop and ground mounted projects. Multiple trainings have also been undertaken under PMKVY scheme, State funding's and other market mode to ensure that skilled workforce are readily made available to meet the requirements of the fast growing Indian solar industry. This Facilitator Guide is designed to enable training for the specific Qualification Pack (QP). The contents of this book are in simple language, without going into too much theoretical details and calculations. It is envisaged that this training manual will provide the participants with the knowledge and skills required for installing and maintaining a Solar Photovoltaic System with a focus on solar rooftop, complying with all applicable codes, standards, and safety requirements; and enable them to actively participate in the growing solar rooftop market.

Units and symbols used in the book have been listed below

Symbols Used



Ask



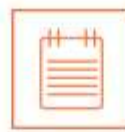
Demonstrate



Facilitation Notes



Learning Outcomes



Notes



Objectives



Practical



Team Activity



Do



Explain



Say



Resources



Activity



Summary



Role Play



Example

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Employability Skills (DGT/VSQ/N0102)

Kindly refer to the Facilitator guide of Employability Skills available on

[Employability skills workbook](#)







1. Introduction to Solar PV Sector in India

UNIT 1.1: Overview of Solar PV Technology

UNIT 1.2: Broader Business, Finance, Technology and Regulatory Landscape of Solar Energy in the Country

UNIT 1.3: Overview of Grid Interactive and Off-Grid Solar Sector in India

UNIT 1.4: Basics of Electrical Concepts

UNIT 1.5: Financial Institutions and Banks Involved in Financing Solar Power Projects



Key Learning Outcomes

At the end of this module, participants will be able to:

1. Discuss and provide overview of solar PV technology, rooftop and ground mounted solar sector in India.
2. Discuss various market research reports and industrial magazines present in the market.
3. Explain different type of ground mounted and Rooftop Solar PV Power Plants and their working principles.
4. Discuss and provide an overview of grid interactive and off- grid Solar Sector in India.
5. Explain type of off- grid Solar PV Power devices and their working principles.
6. Explain various components of a solar PV system and their operating principles.
7. Explain solar energy and power sector landscape in the country.
8. Discuss benefits of solar energy over conventional sources of energy.
9. Explain the broader business, finance, technology & regulatory landscape of solar energy in the country.
10. Analyse the basics of electrical concepts like voltage, current, power, energy, etc
11. Explain typical specifications, functioning, operating principle, maintenance requirements, handling procedures and warranties of different types of solar PV plant components.
12. Demonstrate how to interpret sample Electricity bill of a customer, notices and/or cautions.
13. Explain how to identify various financial institutions and banks involved in lending for solar power projects and analyse their terms & conditions associated with loans.
14. Discuss emerging solar entrepreneurship landscape including success stories of various start- ups.

UNIT 1.1: Overview of Solar PV Technology

Unit Objectives



At the end of this unit, participant will be able to:

1. Discuss and provide overview of solar PV technology, rooftop and ground mounted solar sector in India.
2. Explain different type of ground mounted and Rooftop Solar PV Power Plants and their working principles.
3. Discuss and provide an overview of grid interactive and off- grid Solar Sector in India.
4. Explain type of off- grid Solar PV Power devices and their working principles.
5. Explain various components of a solar PV system and their operating principles.
6. Explain solar energy and power sector landscape in the country.
7. Discuss benefits of solar energy over conventional sources of energy.

Explain



Tell the learners how this presentation will benefit them. Since mature learners are more interested in a subject if they know how or why it is important to them.

Briefly state your credentials in this area, or explain why learners/candidates should listen to you.

Say



Introduce the module to the learners.

Explain



Emphasize on points especially on punctuality and keeping mobile phones switched off. Ensure that the learners understand the importance of adhering to discipline and code of conduct.

Encourage the learners to ask questions for clear understanding and the proper way of asking without interrupting the class.

Additionally, you may tell the learners to also ensure that they are dressed in semi-formal or proper dress code as per the classroom ethics. You may give examples of good code of conduct. For example, somebody whistling in the classroom is not appropriate.

Ask

You may question the learners if they have any idea about the role of solar PV installer in terms of the activities needed to be performed.

Then explain the work flow of solar PV Entrepreneur given here.

Additionally, tell the learners the importance of understanding the work requirement in all aspects before going ahead with the site survey, procurement of components, installation and so on.

Explain

Discuss with the learners the scope of employment opportunities for solar PV Entrepreneurs.

Emphasise the role of this course in skilling the learners to become efficient solar PV Entrepreneurs.

Tell the learners to visit the following site for more information on this:

- <https://www.alliedmarketresearch.com/solar-energy-market>

Discuss the demand of skilled manpower and job opportunities in solar photovoltaics industry in India by 2030.

Inform the learners about different verticals of employment in this industry such as entrepreneur.

Facts and figures given here are as per the report on 'Filling The Skill Gap in India's Clean Energy Market: Solar Energy Focus', by Natural Resources Defense Council (NRDC) and the Council on Energy, Environment and Water (CEEW), February, 2016,

Explain to the learners that key driving forces of this growth of the solar energy market is increase in environmental pollution as well the incentives and tax rebates offered by the government for installation of solar panels.

Explain the career progression graph of solar PV Entrepreneur

Guide the learners how they can improvise and upgrade their skills to progress to a higher level.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Notes for Facilitation

Solutions to Exercise:

Start by giving an introduction about yourself to the learners, using the same points given on the slide.

UNIT 1.2: Broader Business, Finance, Technology and Regulatory Landscape of Solar Energy in the Country

Unit Objectives

At the end of this unit, participant will be able to:

1. Discuss various market research reports and industrial magazines present in the market.
2. Explain the broader business, finance, technology & regulatory landscape of solar energy in the country.
3. Discuss benefits of solar energy over conventional sources of energy.

Explain

Emphasize on various market research reports and industrial magazines present in the market. Explain the key features of the business landscape of solar energy in India.

Encourage the learners to ask questions for clear understanding and the proper way of asking without interrupting the class.

Explain about the various government Scheme incorporated by Government of India to encourage Solar Energy.

Explain the key features of the technology landscape of solar energy in India.

Ask

You may question the learners if they have any idea about the Market research, Finance Landscape, Technology Landscape in Solar PV sector in terms of the activities needed to be performed.

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.

Notes for Facilitation

Three main components of the KUSUM scheme.

Component-A: For Setting up of 10,000 MW of Decentralized Grid Connected Renewable Energy Power Plants on barren land.

Component-B: For Installation of 17.50 Lakh stand-alone solar agriculture pumps

Component-C: For Solarisation of 10 Lakh Grid Connected Agriculture Pumps.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 1.3: Overview of Grid Interactive and Off-Grid Solar Sector in India

Unit Objectives



At the end of this unit, participant will be able to:

1. Discuss and provide an overview of grid interactive and off- grid Solar Sector in India.
2. Explain types of off-grid solar PV power devices and their working principles.

Explain



Explain about the Grid-Interactive Solar Sector and Off-grid Solar Sector. Explain type of off-grid solar PV power devices and their working principles. Explain the working of an off-grid solar water pump.

Ask



You may question the learners if they have any idea about the types of off Grid Solar PV devices in terms of the activities needed to be performed.

Notes for Facilitation



- **Standalone Solar Systems:** These solar power systems were created to supply all the energy requirements of a single home or small company; they are not linked to the main power grid. These systems can be as simple as those that power lighting and mobile devices or as complex as those that can run fans, televisions, and refrigerators.
- **Mini-Grids:** These are distributed solar systems made to supply a collection of homes or small enterprises. Mini-grids, which can supply dependable and reasonably priced electricity to off-grid communities, typically consist of a solar power plant, battery storage, and distribution infrastructure.

UNIT 1.4: Basics Of Electrical Concepts

Unit Objectives

At the end of this unit, participant will be able to:

1. Analyse the basics of electrical concepts like voltage, current, power, energy, etc
2. Explain typical specifications, functioning, operating principle, maintenance requirements, handling procedures and warranties of different types of solar PV plant components.
3. Demonstrate how to interpret sample Electricity bill of a customer, notices and/or cautions.

Explain

Explain the basics of electrical concepts like voltage, current, power, energy, etc. Explain typical specifications of Solar PV Plant Components. Explain the sample copy of electricity bill of any Customer.

Ask

You may question the learners if they have any idea about the Basics of Electrical Concepts devices in terms of the activities needed to be performed.

Demonstrate

Demonstrate how to read and Understand Electricity bill of a customer, notices and/or cautions.

Notes for Facilitation

To read the electricity bill, you need to perform the following tasks:

- Try to capture the bills of at least the 3 most recent months
- Record the units consumed in each month
- Record the unit electricity charges mentioned in the bill

UNIT 1.5: Financial Institutions and Banks Involved in Financing Solar Power Projects

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain how to identify various financial institutions and banks involved in lending for solar power projects and analyse their terms & conditions associated with loans.
2. Discuss emerging solar entrepreneurship landscape including success stories of various start-ups.

Explain

Explain the prominent financial institutions and banks involved in lending solar power projects. Explain the terms and conditions associated with these loans. Explain stories of various start-ups emerging in solar energy entrepreneurship.

Ask

You may question the learners if they have any idea about the Financing in Solar Power projects in terms of the activities needed to be performed.

Demonstrate

Demonstrate how to read and understand the terms associated with lending of Solar Power Projects.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.





2. Set up new venture for Solar Photovoltaic System

UNIT 2.1: Setting Up a New Venture

UNIT 2.2: Workplace Etiquette

UNIT 2.3: Time Management, Leadership Skills and Resource Management

UNIT 2.4: Productivity Tools: MS Word, MS Excel, PowerPoint and Presentations

UNIT 2.5: Challenges and Risks for New Entrepreneurs

UNIT 2.6: Taxes and Compliances

UNIT 2.7: Aspects of Successful Solar Business

UNIT 2.8: Respecting Gender and Disability Concerns



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Describe the process for setting up a new venture including required registrations and availing low-cost financing from Financing Institutions (FIs) and Venture Capital (VC) firms.
2. Discuss various evolving opportunities across solar solutions, services and market ecosystem.
3. Discuss how to identify the key ingredients of a business plan.
4. Discuss how to distinguish between fixed and working capital requirements.
5. Describe the components of a loan application.
6. Discuss how to demonstrate good etiquettes and manners while communicating with the client.
7. Discuss how to demonstrate the importance of time management.
8. Discuss how to demonstrate leadership skills and effective resource management techniques.
9. Discuss the use of MS Word and Excel for preparing a business proposal
10. Discuss to prepare a workable presentation for marketing and business development.
11. Discuss how to choose the right buyer in a given situation of market parameters.
12. Identify the challenges and risks for new entrepreneurs and the possible mitigation measures.
13. Explain various taxation and related compliances to set up and operate a solar business
14. Discuss the key aspects to successfully run a solar business.
15. Explain how evolving Start-ups across solar Consultancy/O&M/Installation and EPC space, and diversifying solar portfolio is improving overall solar landscape.
16. Explain how to demonstrate responsible and disciplined behaviour and promote a safe and interactive environment at work while respecting gender and disability concerns.

UNIT 2.1: Setting Up a New Venture

Unit Objectives

At the end of this unit, participant will be able to:

- Describe the process for setting up a new venture including required registrations and availing low-cost financing from Financing Institutions (FIs) and Venture Capital (VC) firms.
- Discuss various evolving opportunities across solar solutions, services and market ecosystem.
- Discuss how to identify the key ingredients of a business plan.
- Discuss how to distinguish between fixed and working capital requirements.
- Describe the components of a loan application.

Explain

Briefly introduce about entrepreneurship to the learners. Tell the Learners about the essential qualities for entrepreneurs.

Let them know the Basic Steps Involved in Setting Up a New Venture.

Inform the learners about the Types of Organisations and Required Registration.

Explain the image given on the slide by elaborating the Generic structure of a solar rooftop business model.

Do

Prepare List of Some Common Components of Loan Application required for a Solar PV Project.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 2.2: Workplace Etiquette

Unit Objectives

At the end of this unit, participant will be able to:

- Discuss how to demonstrate good etiquettes and manners while communicating with the client

Explain

Explain to the learners about Forms of Communication as given on the slide. Explain some language and listening barriers in communication. Tell the learners about how to Interact with Others without Any Bias.

Tell the Learners to Interacting Effectively with Clients & Colleagues

Learners must know about Recording and Forwarding Accurate Information in the organization.

Demonstrate

Demonstrate good etiquettes and manners while communicating with the client.

Say

Read out the points given on the slides.

You may also show videos related to conflicts that arise due to perception barriers.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 2.3: Time Management, Leadership Skills and Resource Management

Unit Objectives

At the end of this unit, participant will be able to:

- Discuss how to demonstrate the importance of time management.
- Discuss how to demonstrate leadership skills and effective resource management techniques.

Explain

Tell the learners about benefits of Time Management.

Explain effective Time Management Techniques. Explain Leadership Qualities That All Entrepreneurs Need. Explain Benefits of Effective Leadership. Explain about Resource Management.

Ask

You may question the learners if they have any idea about the Time Management/ Resource Management in terms of the activities needed to be performed.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught

Notes for Facilitation

Benefits of Effective Leadership

Effective leadership results in numerous benefits. Great leadership leads to the leader successfully:

- Gaining the loyalty and commitment of the team members
- Motivating the team to work towards achieving the company's goals and objectives
- Building morale and instilling confidence in the team members
- Fostering mutual understanding and team-spirit among team members
- Convincing team members about the need to change when a situation requires adaptability

UNIT 2.4: Productivity Tools: MS Word, MS Excel, PowerPoint and Presentations

Unit Objectives

At the end of this unit, participant will be able to:

- Discuss the use of MS Word and Excel for preparing a business proposal
- Discuss to prepare a workable presentation for marketing and business development.

Explain

Tell the learners about Microsoft Word, Microsoft Excel, PowerPoint.

Explain how to launch them and how to create document and Presentations. Explain how to make bar chart and tables, pictures and graphs. How to save the document.

Ask

You may question the learners if they have any idea about the Microsoft office tools in terms of the activities needed to be performed.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught

Do

Use MS Word and MS Excel and its applications to prepare business proposal that should contain the following details:

- Area available
- Partner's Details
- Sanction of Loan (if any) by financial institution
- Machinery
- Recruitment of technical team
- Provision of facilities like water, electricity etc.
- License details
- Compliance letter

UNIT 2.5: Challenges and Risks for New Entrepreneurs

Unit Objectives

At the end of this unit, the participant will be able to:

- Identify the challenges and risks for new entrepreneurs and the possible mitigation measures.
- Discuss how to choose the right buyer in a given situation of market parameters.

Explain

Tell the learner about Common Challenges and Risks for New Entrepreneurs, along With Possible Mitigation Measures. Explain steps to consider when selecting the right buyer in a given situation of market parameters.

Ask

You may question the learners if they have any idea about the Challenges and Risks in terms of the activities needed to be performed.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.

UNIT 2.6: Taxes and Compliances

Unit Objectives

At the end of this unit, the participant will be able to:

- Explain various taxation and related compliances to set up and operate a solar business

Explain

Tell the learner about various taxation and related compliances that a solar business must comply with. Explain Goods and services Tax. Explain Renewable Purchase Obligations (RPO).

Ask

You may question the learners if they have any idea about the Challenges and Risks in terms of the activities needed to be performed.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.

UNIT 2.7: Aspects of Successful Solar Business

Unit Objectives

At the end of this unit, the participant will be able to:

- Discuss the key aspects to successfully run a solar business.
- Explain how evolving Start-ups across solar Consultancy/O&M/Installation and EPC space, and diversifying solar portfolio is improving overall solar landscape.

Explain

Tell the learner about Key Aspects to Successfully Run a Solar Business. Explain Impact of Evolving Start-ups across solar Consultancy/O&M/Installation and EPC space, and diversifying solar portfolio in improving overall solar landscape.

Ask

You may question the learners if they have any idea about the Challenges and Risks in terms of the activities needed to be performed.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

Do

Watch the Video to know more about Key Aspects to Successfully Run a Solar Business, you may refer to the following link:

<https://www.youtube.com/watch?v=W3QuS47UrpQ>

UNIT 2.8: Respecting Gender and Disability Concerns

Unit Objectives

At the end of this unit, the participant will be able to:

- Explain how to demonstrate responsible and disciplined behaviour and promote a safe and interactive environment at work while respecting gender and disability concerns.

Explain

Tell the learner about Key elements of Diversity and Inclusion.

Explain Ways to Improve the Environment and Culture of Organisation. Explain the definition of Gender as per UNESCO and Explain Schemes by Government of India to address PwD sensitivity issues.

Ask

You may question the learners if they have any idea about the Challenges and Risks in terms of the activities needed to be performed.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

Notes for Facilitation

General points to be kept in mind:

- Do use a common language understood by all while speaking to a group.
- Do speak to everyone with respect in the workspace, interact with all co-workers without discrimination.
- Do not bully or make anyone feel inferior or left out.
- Do have open discussions and involve everyone.
- If possible, conduct informal get-togethers, celebrate festivals. It will help in engaging every employee, improving the organization culture and making the employees feel appreciated.



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain the requirements for safe work area.
2. Explain the importance of administering first aid.
3. Identify the personal protective equipment used for the specific purpose.
4. Identify the hazards associated with photovoltaic installations;
5. Identify work safety procedures and instructions for working at height.
6. Explain the importance of Occupational health & Safety standards and regulations for installation of Solar PV system.
7. Incorporate good housekeeping practices and infection control guidelines.

UNIT 3.1: Workplace Safety

Unit Objectives

At the end of this unit, participant will be able to:

1. Recognize the potential sources of accidents
2. Classify the safety policies
3. Identify the responsibilities of managers and employees regarding safety
4. List the general safety guidelines
5. Recognize the safety measures and precautions

Say

Introduce the module name to the learners and the relevance of maintaining health and safety standards at the project site.

Notes for Facilitation

Tell them the importance of keeping workplace safe and clean.

Inform them about how to communicate the issues and problems to the higher authorities.

Explain

Explain the types of hazards that a solar technician may face and the reasons of such hazards.

Ask

Ask them to protect themselves of such hazards by wearing PPE, when required

Notes for Facilitation

Discuss some measures with them they need to take in case of any mishappening.

Demonstrate

Demonstrate the correct method of giving first-aid. Tell them the importance of giving first-aid before the arrival of medical help.

UNIT 3.2: Workplace Hazards

Unit Objectives



At the end of this unit, participant will be able to:

1. List different hazards present at the workplace.
2. Identify the safety requirements at workplace.
3. List the precautionary requirements for health at workplace.

Unit Objectives



At the end of this unit, you will be able to:

1. Identify environmental hazards associated with photovoltaic installations
2. Identify electrical hazards
3. Identify personal safety hazards or work site hazards and Mitigate hazards

Say



Introduce the concept of 'hazard' to them. Tell them the examples of each types of hazard.

Personal safety hazards are caused due to humans' negligence.

Electrical hazards are caused due to electrical issues such as short circuit, voltage fluctuations, damaged machinery etc.

Non-electrical hazards are other hazards which are not caused due to any electronic component.

Environmental hazards are caused because of environmental factors such as pollution, gas leak etc.

Notes for Facilitation



Discuss some examples of electrical hazards with them such as battery hazards, inverter hazards and AC power hazards.

Demonstrate



Demonstrate the procedure of operating a fire extinguisher to them.

Tell them that it is important to keep yourself safe while using fire extinguisher.

UNIT 3.3: Standard Safety Practices

Unit Objectives

At the end of this unit, participant will be able to:

1. List the personal protection practices.
2. Identify the standard safety practices at workplace.
3. Explain the improvement opportunities at workplace.

Resources to be Used

Show them the constituents of PPE.

Tell them to wear PPE only when required and replace the damaged one.

Explain

Explain the use of every equipment to them such as:

- Helmet will help them to keep their head safe from an unexpected fall.
- Shoes will help them to prevent any slippage.
- Any ring should not be worn on a finger to prevent getting stuck with the object.
- Wear goggles in case of wiring work.
- Hand and arm protection is needed to keep one safe from fire spark.

Notes for Facilitation

Body harness is needed when a solar technician is working on a height and needs protection from falling.

Safety jacket is worn so that one can remain visible from a distance.

UNIT 3.4: Safety Actions

Unit Objectives

At the end of this unit, participant will be able to:

1. List steps of first aid.
2. Prepare a report of periodic checks.
3. Explain how to provide help to heart attack victims.

Explain

Explain the basic techniques of Bandaging. Explain First Aid to Victims of Heart Attack
Tell them steps for performing the CPR.

Ask

Ask the learners to attempt the questions. Help them if required.

Do

Assist the learners to report the periodic checks. Show them the benefits of good housekeeping at workplace.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

Notes for Facilitation

The steps for performing the CPR are as follows:

Step 1: Check for pulse on the wrist and the neck.

Step 2: Check for breathing and see if the person is conscious.

Step 3: Ask someone to call for medical help.

Step 4: Position the victim, making sure that the person's face is upwards. If the person has an injury on the head or the neck, aids must be provided without moving him/her.

Step 5: Apply pressure on the centre of the chest.



4. Market Research and Cost Estimation of Rooftop

Unit 4.1 – Assess the Market Trends in Rooftop Solar

Unit 4.2 – Assess the Quality of Solar Components

Unit 4.3 – Assess the Cost of Solar PV Power Plant

Unit 4.4 – Select the Business Model for Solar Rooftop Sector



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to assess the market trends and evaluate the business prospect for rooftop solar.
2. Discuss how to identify market opportunities and potential customers.
3. Explain how to select the right quality of solar module by identifying the key technical parameters in data specification sheets.
4. Discuss how to select the right quality of inverter by identifying the key technical parameters in data specification sheets.
5. Discuss how to select the right quality of mounting structure by identifying the key technical parameters in data specification sheets.
6. Discuss how to select the right quality of battery by identifying the key technical parameters in data specification sheets.
7. Discuss how to identify market price of different components of solar PV system.
8. Discuss how to prepare cost estimate for a rooftop solar project.
9. Explain to prepare a cost benefit analysis for a rooftop solar PV plant including Levelised Cost of Energy (LCOE), Payback, Internal Rate of Return (IRR) etc.
10. Discuss to identify the policy, regulations and procedures for solar rooftop sector in the local market.
11. Explain to identify and select the appropriate business models in solar rooftop sector.

UNIT 4.1: Assess the Market Trends in Rooftop Solar

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to assess the market trends and evaluate the business prospect for rooftop solar.
- Discuss how to identify market opportunities and potential customers.

Explain

Explain various processes involved in analysing market trends and business opportunities for rooftop solar. Explain Ways to Identify Market Opportunities and Potential Customers.

Ask

You may question the learners if they have any idea about the Market Trends and Market Opportunities in terms of the activities needed to be performed.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.

UNIT 4.2: Assess the Quality of Solar Components

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to select the right quality of solar module by identifying the key technical parameters in data specification sheets.
- Discuss how to select the right quality of inverter by identifying the key technical parameters in data specification sheets.
- Discuss how to select the right quality of mounting structure by identifying the key technical parameters in data specification sheets.
- Discuss how to select the right quality of battery by identifying the key technical parameters in data specification sheets

Explain

Explain Procedure for Selecting the Right Quality of Solar Module. Explain Procedure for Selecting the Right Quality of Solar Module. Explain Procedure for Selecting the Right Quality of Inverter, Battery, Mounting Structure.

Ask

You may question the learners if they have any idea about the Market Trends and Market Opportunities in terms of the activities needed to be performed.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Demonstrate

Demonstrate the Sample Data Specification Sheet for Solar PV Panel, Battery, Inverter.

UNIT 4.3: Assess the Cost of Solar PV Power Plant

Unit Objectives

At the end of this unit, participant will be able to:

- Discuss how to identify market price of different components of solar PV system.
- Discuss how to prepare cost estimate for a rooftop solar project.
- Explain how to prepare a cost benefit analysis for a rooftop solar PV plant including Levelised Cost of Energy (LCOE), Payback, Internal Rate of Return (IRR) etc.

Explain

Explain Factors Impacting the Price of Different Components of a Solar PV System in India. Explain Key Steps to Prepare Cost Estimate for a Rooftop Solar Project. Explain how to prepare a cost-benefit analysis for a rooftop solar PV plant. Explain the Procedure to calculate the Levelized Cost of Energy (LCOE)

Ask

You may question the learners if they have any idea about the Cost of Different components of solar PV system in terms of the activities needed to be performed.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Demonstrate

Demonstrate the formula to calculate the Net Present Value (NPV) of a project or investment.

Notes for Facilitation

- $IRR = r1 + ((NPV1 / (NPV1 - NPV2)) \times (r2 - r1))$

$$NPV = PV \text{ (Cash inflows)} - PV \text{ (Cash outflows)}$$

- $PV = \text{Cash flow} / (1 + r)^n$

UNIT 4.4: Select the Business Model for Solar Rooftop Sector

Unit Objectives

At the end of this unit, participant will be able to:

- Identify the policy, regulations and procedures for solar rooftop sector in the local market.
- Identify and select the appropriate business models in solar rooftop sector.

Explain

Explain various policies, regulations, and procedures to support the growth of the solar rooftop sector. Explain Renewable Energy Service Company (RESCO) Model. Explain Community Solar Model.

Ask

You may question the learners if they have any idea about the Business Model for Solar Rooftop Sector in terms of the activities needed to be performed.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Demonstrate

Demonstrate RESCO Model and Community Solar Model.



5. Site Feasibility Study (Rooftop Solar)

Unit 5.1 - Undertaking Site Feasibility

Unit 5.2 - Perform Shading Analysis using Various Tools

Unit 5.3 - Identify Risks with Various Solar Projects



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Identify optimum location of rooftop solar plants installations
2. Assess the site level pre-requisites for solar panel installation
3. Analyse type of mounting to be constructed and place of mounting as per client requirement
4. Check for any shading obstacles
5. Prepare a site map of the location where installation has to be carried out
6. Assess the load to be run on solar PV power plant and prepare a load profile
7. Estimate the capacity of solar PV power plant
8. Discuss on how to decide on battery backup as per grid availability, loads and client expectation
9. Assess or obtain the site-specific major parameters of solar resource data like GHI, DNI, Temperature and Wind
10. Perform shading analysis
11. Estimate the energy generated from the rooftop solar PV power plant using solar design software like PV*SOL®, PVsyst, etc.
12. Identify the risks associated with the specific solar project
13. Prepare a site Feasibility Study Report using specialized software like PV*SOL®, PVsyst, etc.

UNIT 5.1: Undertaking Site Feasibility

Unit Objectives



At the end of this unit, participant will be able to:

- Identify optimum location of rooftop solar plants installations
- Assess the site level pre-requisites for solar panel installation
- Analyse type of mounting to be constructed and place of mounting as per client requirement
- Prepare a site map of the location where installation has to be carried out

Explain



Explain different types roofs. Explain the information available from site feasibility report. Explain will help in identifying an optimum location for installation. Explain different types of Mountings. Explain the type of information required while creating a site map.

Ask



You may question the learners if they have any idea about the site feasibility for Solar Rooftop Sector in terms of the activities needed to be performed.

Say



Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Demonstrate



Calculate Area required by solar PV plants either for rooftop or ground mounted

Notes for Facilitation



- The area required for PV panel installation = Total number of solar panels
 - The size of one 330-watt PV panel with 72 cell configuration = 3.25 ft X 6.42 ft.
 - The surface area of one 330-watt solar panel = 21.51 sq. ft.
 - A 7.5 kW system has 23 solar panels each of 330 watts, which means area needed to install 7.5 kW of solar panels is:
 - Area for 7.5 kW of solar panels = 23 x 21.50 = 495 sq. ft

UNIT 5.2: Perform Shading Analysis using Various Tools

Unit Objectives

At the end of this unit, participant will be able to:

1. Perform shading analysis
2. Check for any shading obstacles
3. Assess or obtain the site-specific major parameters of solar resource data like GHI, DNI, Temperature and Wind
4. Prepare a site Feasibility Study Report using specialized software like PV*SOL®, PVsyst, etc.
5. Estimate the energy generated from the rooftop solar PV power plant using solar design software like PV*SOL®, PVsyst, etc.
6. Estimate the capacity of solar PV power plant
7. Assess the load to be run on solar PV power plant and prepare a load profile

Explain

Explain role of shading analysis in solar PV System. Explain tools used in shading analysis of Solar PV System. Explain Solar Irradiance. Explain different components of solar Irradiance.

Ask

You may question the learners if they have any idea about the shading analysis for Solar Rooftop Sector in terms of the activities needed to be performed.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Demonstrate

Calculate Battery Capacity, Inverter Capacity, Solar Panel Capacity and Calculate Electric Load on Solar PV Panels.

Notes for Facilitation

- Global Horizontal Irradiance (GHI) = Direct Normal Irradiance (DNI)* cos (solar zenith angle) + Diffused Horizontal Irradiance (DHI)

Load per day = Rating X Number of operation hours per day

UNIT 5.3: Identify Risks with Various Solar Projects

Unit Objectives

At the end of this unit, participant will be able to:

1. Identify the risks associated with the specific solar project.
2. Discuss on how to decide on battery backup as per grid availability, loads and client expectation

Explain

Explain Risk associated with various solar Projects. Explain the step-by-step procedure to perform a load and site assessment for any customer

Ask

You may question the learners if they have any idea about the Risk associated with Solar Rooftop Sector in terms of the activities needed to be performed.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Notes for Facilitation

Procedure to Read the Electricity Bill

- Try to capture the bills of at least the 3 most recent months
- Record the units consumed in each month
- Record the unit electricity charges mentioned in the bill
- Also, note the additional surcharges that have been applied by the distribution company. This will be useful when later explaining the payback to the consumer.

To know more about How to read Electricity Bill, you may refer to the following link:

<https://www.youtube.com/watch?v=qIBvhNKd3JY>



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to oversee to cast the Civil foundation (Footings/Pedestal) on roof
2. Ensure the electrical installation tools shall be available on site
3. Discuss how to oversee the process of installing Mounting structure.
4. Discuss how to install the foundation for ACDB, Inverter etc
5. Explain how to install the Energy meter
6. Discuss how to install the AC isolator
7. Discuss how to install the Earthing and Lightning arrestor on site.
8. Discuss the know-how of electrical installation tools.
9. Explain the process of testing and commissioning of equipment in DC and AC side of the plant.

UNIT 6.1: Cast Civil foundation

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to oversee to cast the Civil foundation (Footing/Pedestal) on roof

Explain

Explain the image shown on the slide.

Tell the learners about the different types of footing. Explain tools and consumables required for Civil/Mechanical Installation. Explain Installation of mounting post for fixed structures with legs

Ask

You may question the learners if they have any idea about the Civil foundation for Solar Rooftop Sector in terms of the activities needed to be performed.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Demonstrate

Sample PV Array Layout Design, Important tools and consumables required to do civil/mechanical work, Sample design for installing mounting post.

UNIT 6.2: Preparing Electrical Design of Solar PV Power Plant

Unit Objectives



At the end of this unit, participant will be able to:

1. Ensure the electrical installation tools shall be available on site
2. Discuss the know-how of electrical installation tools.
3. Discuss how to install the foundation for ACDB, Inverter etc
4. Explain how to install the Energy meter
5. Discuss how to install the AC isolator
6. Discuss how to install the Earthing and Lighting arrestor on site.
7. Explain the process of testing and commissioning of equipment in DC and AC side of the plant.

Explain



Explain the Tools and Tackles Required for Solar PV Installation. Explain how to Select the Location of DC Combiner Box. Explain the steps for Installation of DC combiner box. Explain steps to install inverter

Say



Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Notes for Facilitation



Mounting of inverters included following steps:

- First of all check the dimensions and knockout locations.
- Put bracket on the wall and mark the position of holes by using a pencil.
- Drill the hole locations by using drill machine.
- Mount the bracket to the wall using the screws.
- Ensure that screws are fully tight.
- Make sure that there is adequate clearance around the
- inverter for secure and optimal working.

To know more about Solar Inverter Installation, you may refer to the following link:

<https://www.youtube.com/watch?v=2D8bMRZ280I>

UNIT 6.3: Earthing of the PV system

Unit Objectives



At the end of this unit, participant will be able to:

1. Explain different types of Earthing systems.
2. Discuss the Earthing Design.
3. Discuss Rules for grounding solar PV system.

Explain



Tell the Learners about earthing and different types of earthing systems. Explain sample block diagram of earthing of a solar PV plant given on slide. Explain rules for grounding of PV system

Say



Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Notes for Facilitation



As per the safety standards, it is mandatory that every solar plant should have earthing/ grounding system in order to:

1. Safeguard the solar PV plant from lightning.
2. Protect other nearby objects, humans, animals from any electric shock.
3. Avoid electric faults.
4. Maintain Solar PV plant stability.
5. Maintain and enhance performance of devices attached to the Solar PV plant from any lightning conductor system that surrounds them
6. Solar panels have in built transformers that set up the voltage generated through solar power plants. Due to their designs, they are more prone to shocks. In case you require earthing for a PV solar power system, you need an earth cable, earthing joint, and an earth plate.
7. Protect the workers around the solar power plants.
8. Keep solar power plants in check at regular intervals.
9. A grounding path will easily avoid any electrical fault and would not pose a threat to system stability.
10. Have a better functionality of devices.

To know more about Earthing of Solar PV Systems, you may refer to the following link:

<https://www.youtube.com/watch?v=GQMgMZn3WBU>

UNIT 6.4: Testing of the PV system

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain the Procedure for Voltage testing of solar Panel.
2. Explain the Procedure for Current testing of solar Panel.
3. Discuss how to test the various electrical components like, battery, Load LED, Load wire and PCB.

Explain

Tell the Learners about Current testing, Voltage Testing of Solar Panel. Explain the procedure to test the various electrical components like, battery, Load LED, Load wire and PCB.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Notes for Facilitation

Technical Testing: LED

Step 1: Place the black probe on the negative terminal of the LED

Step 2: Place the red probe on the positive terminal of the LED

Step 3: Apply 5V DC current

Step 4: If the LED glows, accept the LED. If it does not glow reject the LED

To know more about testing of solar PV system, you may refer to the following link:

<https://www.youtube.com/watch?v=eh6zMb69gY>

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.



7. Operation and Maintenance of Solar PV Rooftop Plant

UNIT 7.1: Tools and Equipment for O&M

UNIT 7.2: Working Condition of Various Components

UNIT 7.3: Implement approaches for O&M



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to handle and utilise tools and equipment for O&M available at site.
2. Explain how to check the integrity and working condition of all connections, fuses and circuit breakers.
3. Ensure working condition of the junction boxes/combiner boxes.
4. Ensure the continuity of cables and wires are maintained.
5. Ensure proper electrical connections throughout the solar PV power plant up to the inverter input.
6. Discuss how to verify the connections, cables and junction boxes as per the design/working drawings.
7. Explain how to troubleshoot the identified faults and escalate the issue to concerned person for timely resolution.
8. Explain how to verify the earthing and lightening protection systems as per the as-built drawings and report in case of any discrepancies.
9. Ensure proper cleaning of modules as per schedule and standard procedure and remove any shadowing objects.
10. Discuss how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance.
11. Ensure that the system is well maintained and is performing at an optimum level.
12. Discuss best practices and successful case studies on performing O&M.

UNIT 7.1: Tools and Equipment for O&M

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain how to handle and utilise tools and equipment for O&M available at site.

Explain

Tell the Learners about List of tools and materials required for O&M of solar systems. Explain about Maintenance schedule and logbook.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Notes for Facilitation

- An equipment logbook incorporating the maintenance schedules will be provided to the technicians
- A loose-leaf folder can be used as the system log book with individual sheets added for each item.
- Log books are useful because the historical information they contain can show changes over time, as well as abnormal variations from the usual, indicating a problem in the making.

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.

UNIT 7.2: Working Condition of Various Components

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to check the integrity and working condition of all connections, fuses and circuit breakers.
- Ensure working condition of the junction boxes/combiner boxes.
- Ensure the continuity of cables and wires are maintained.
- Ensure proper electrical connections throughout the solar PV power plant up to the inverter input.
- Discuss how to verify the connections, cables and junction boxes as per the design/working drawings.

Explain

Tell the Learners about Process of checking Continuity of Cables and Wires. Explain the procedure of finding a ground fault. Explain how to Inspect Physically damaged wires.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Notes for Facilitation

- Visually check all conduit and wire insulation for damage.
- Check for loose, broken, corroded, or burnt wiring connections.
- Check if all equipments are connected with proper wire and conduit.
- Make sure all wiring is secured, by gently but firmly pulling on all connections.
- Check all terminals and wires for loose, broken, corroded, or burnt connections or components
- Check the operation of the PV array DC isolators
- Check array wiring for physical damage and wind chafing
- Check array mounting hardware for tightness and corrosion

Inspection and maintenance of Earthing and Lightning Protection:

- Use an ohmmeter or multimeter to check the continuity of the entire grounding system.
- Make sure that all module frames, metal conduit and connectors, junction boxes, and electrical components chassis are earth grounded.

To know more about Testing of cable/ wire, you may refer to the following link:

<https://www.youtube.com/watch?v=cgqzezYaYkc>

UNIT 7.3: Implement approaches for O&M

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain how to troubleshoot the identified faults and escalate the issue to concerned person for timely resolution.
2. Explain how to verify the earthing and lightening protection systems as per the as-built drawings and report in case of any discrepancies.
3. Ensure proper cleaning of modules as per schedule and standard procedure and remove any shadowing objects.
4. Discuss how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance.
5. Ensure that the system is well maintained and is performing at an optimum level.
6. Discuss best practices and successful case studies on performing O&M.

Explain

Tell the Learners about how to troubleshoot faults in a solar PV system. Explain Earthing/Grounding of a Solar PV System. Explain the procedure of cleaning solar PV Panels.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Notes for Facilitation

- How to clean PV modules?
 - Inspect the site prior to cleaning work is performed. Note any hazardous conditions and danger.
 - Cover all electrical equipment, i.e. inverters and combiner boxes, beneath area to be cleaned with covers prior to cleaning.
 - Inspect surface of solar modules prior to cleaning. All accumulated dirt to be removed from module surface.
 - Use ladder wherever required to reach surface of solar modules to be cleaned.
 - Use a wiper with a sponge or brush to clean modules. Use clean water at minimal pressure. No chemicals or abrasive cleaning shall be applied to clean solar modules.
- To know more about Solar Operations and Maintenance, you may refer to the following link:
<https://www.youtube.com/watch?v=pmWlvUWkeQo>





8. Manage Solar PV Project Lifecycle

UNIT 8.1: Read and Interpret the Diagram and Drawings

UNIT 8.2: Prepare Lifecycle Cost of a Rooftop Solar Project

UNIT 8.3: Analysing Rooftop Solar Project

UNIT 8.4: Maintain a Rooftop Solar PV Power Plant



Key Learning Outcomes

At the end of this module, the participant will be able to:

- Discuss how to read and interpret the single line diagram, civil / mechanical drawings and electrical drawings.
- Explain how to read and interpret the bill of material.
- Explain how to calculate the lifecycle cost of a rooftop solar project.
- Identify and mitigate various risks associated with the project.
- Discuss how to ensure the solar PV system and structure meets the local government and regulatory requirements.
- Prepare action plan and coordinate the implementation of rooftop solar project.
- Discuss how to identify the maintenance activity required for a rooftop solar PV power plant component.
- Prepare a preventive maintenance schedule.
- Ensure proper cleaning of solar panels periodically.

Ensure regular inspection of the solar PV system to identify and rectify the faults.

UNIT 8.1: Read and Interpret the Diagram and Drawings

Unit Objectives

At the end of this unit, participant will be able to:

- Discuss how to read and interpret the single line diagram, civil / mechanical drawings and electrical drawings.
- Explain how to read and interpret the bill of material.

Explain

Tell the Learners about Lifecycle of a Solar PV (Photovoltaic) project. Explain electrical design of the Solar PV system. Explain mechanical design of solar PV system. Explain sample bill of material.

Resources to be Used

Show them the constituents of PPE.

Tell them to wear PPE only when required and replace the damaged one.

Demonstrate

Demonstrate Sample Electrical Design of Solar PV Panel.

Demonstrate the Single Line diagram for a 50KW solar PV system as given in Module.

Demonstrate Sample Bill of Materials.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 8.2: Prepare Lifecycle Cost of a Rooftop Solar Project

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to calculate the lifecycle cost of a rooftop solar project.

Explain

Explain how to calculate lifecycle cost of solar PV Project. Explain the procedure for calculating net present value of savings.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

Notes for Facilitation

To calculate the Net Present Value of Savings, the following formula can be used:

Net Present Value of Savings = $(\text{Annual Savings} / \text{Discount Rate}) \times ((1 - (1 + \text{Discount Rate})^{-\text{System Lifespan}}) / \text{Discount Rate})$

where:

Annual Savings: This is the annual savings generated by the solar PV system, which is the difference between the energy generated by the system and the building's historical energy consumption.

Discount Rate: This is the discount rate used to discount the future cash flows to their present value. This rate reflects the time value of money and the risk associated with the investment.

System Lifespan: This is the expected lifespan of the solar PV system in years.

UNIT 8.3: Analysing Rooftop Solar Project

Unit Objectives

At the end of this unit, participant will be able to:

- Identify and mitigate various risks associated with the project.
- Discuss how to ensure the solar PV system and structure meets the local government and regulatory requirements.
- Prepare action plan and coordinate the implementation of rooftop solar project.

Explain

Explain Common Risks Associated with Solar PV Projects. Explain Local Government and Regulatory Requirements for Solar PV System and Structure. Explain Key Steps Involved in Coordinating the Implementation of a Rooftop Solar Project.

Notes for Facilitation

Tell them that it is important to consider these steps in implementation of a rooftop solar project

- **Assign Roles and Responsibilities:** Identify team members and assign specific roles and responsibilities for the project. This ensures that everyone knows what is expected of them and can work towards the same goals.
- **Develop a Timeline:** Create a timeline for the project that includes specific milestones and deadlines. This will help to keep the project on track and ensure that everyone is working towards the same deadlines.
- **Establish Communication Channels:** Establish communication channels between team members, stakeholders, and vendors to ensure effective communication throughout the project.
- **Monitor Progress:** Regularly monitor progress against the timeline and adjust as necessary. This will help to identify potential issues early and allow for corrective action to be taken.
- **Manage Resources:** Ensure that resources, such as materials and equipment, are available when needed and that they are allocated efficiently.
- **Maintain Documentation:** Maintain documentation throughout the project to track progress, identify issues, and communicate with stakeholder

UNIT 8.4: Maintain a Rooftop Solar PV Power Plant

Unit Objectives

At the end of this unit, participant will be able to:

- Discuss how to identify the maintenance activity required for a rooftop solar PV power plant component.
- Prepare a preventive maintenance schedule.
- Ensure proper cleaning of solar panels periodically.
- Ensure regular inspection of the solar PV system to identify and rectify the faults.

Explain

Tell the learners about the maintenance activity required for a rooftop solar PV power.

Explain the Procedure to create the preventive maintenance schedule.

Tell the learners to be complaint with the ideal practice of IEEE standard of bonding electrical equipment.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Notes for Facilitation

A preventive maintenance plan is essential for the optimal performance, safety, and longevity of a rooftop solar PV plant. By implementing a regular maintenance schedule, potential issues can be identified and addressed before they become significant problems, ensuring the system operates at its optimal level and lasts for many years.

To know more about Troubleshooting of solar PV Systems, you may refer to the following link:
https://www.youtube.com/watch?v=MLNVA_iusAs

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.





Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to assess the market trends and evaluate the business prospect for solar water pumping
 2. Discuss how to identify market opportunities and potential customers.
 3. Explain how to select the right quality of solar module by identifying the key technical parameters in data specification sheets
 4. Discuss how to select the right quality of material and mounting structure by identifying the key technical parameters in data specification sheets
 5. Discuss how to select the right quality of battery by identifying the key technical parameters in data specification sheets
 6. Discuss how to identify market price of different components of solar water pump
 7. Discuss how to prepare cost estimate for a solar water pump.
 8. Explain to prepare a cost benefit analysis for a solar water pumping including Levelised Cost of Energy (LCOE), Payback, Internal Rate of Return (IRR) etc.
 9. Discuss to identify the policy, regulations and procedures for deployment of solar water pumping system.
- Explain to identify and select the appropriate business models in solar water pumping.

UNIT 9.1: Assess the Market Trends in Solar Pumping Business

Unit Objectives



At the end of this unit, participant will be able to:

- Explain how to assess the market trends and evaluate the business prospect for solar water pumping
- Discuss how to identify market opportunities and potential customers.

Explain



Introduce the module to the learners.

Emphasize that an important aspect of Solar PV Water Pumping System Market Overview.

Explain SWOT analysis for Solar PV Water Pumping business in India given in module.

Explain Ways to Identify Market Opportunities and Potential Customers

Notes for Facilitation



Marketing Plan

Develop a marketing plan to promote the solar water pumping business. Determine the marketing channels, such as social media, online advertising, and direct marketing, and develop a messaging strategy. Your Marketing plan should include the following strategies:

- Market penetration strategy: This strategy focuses on selling your existing products or services in existing markets, in order to increase your market share.
- Growth strategy: This strategy focuses on increasing the amount of market share, even if it reduces earnings in the short-term.
- Channels of distribution strategy: These can be wholesalers, retailers, distributors and even the internet.
- Communication strategy: These can be written strategies (e-mail, text, chat), oral strategies (phone calls, video chats, face-to-face conversations), non-verbal strategies (body language, facial expressions, tone of voice) and visual strategies (signs, webpages, illustrations).

Role Play



Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.

UNIT 9.2: Assessing the Quality of Solar Component for Solar Pumps

Unit Objectives



At the end of this unit, participant will be able to:

- Explain how to select the right quality of solar module by identifying the key technical parameters in data specification sheets
- Discuss how to select the right quality of material and mounting structure by identifying the key technical parameters in data specification sheets
- Discuss how to select the right quality of battery by identifying the key technical parameters in data specification sheets

Explain



Introduce the module to the learners. Explain the procedure for Selecting Right Quality of Components. Explain Data Specification Sheet for Solar PV Panel Required for Solar PV Water Pump. how to select the right quality of battery by identifying the key technical parameters in data specification sheets

Notes for Facilitation



Selecting the Right Quality of Inverter

Following key specification (but not limited to these) must be taken into consideration for procuring the Inverters.

- Type of Inverter
- AC Output capacity
- DC input voltage range
- Operating temperature range

Role Play



Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.

UNIT 9.3: Assess the Cost of Solar Water Pump and Carry out Basic Financial Calculations

Unit Objectives



At the end of this unit, participant will be able to:

- Discuss how to identify market price of different components of solar water pump
- Discuss how to prepare cost estimate for a solar water pump.
- Explain to prepare a cost benefit analysis for a solar water pumping including Levelised Cost of Energy (LCOE), Payback, Internal Rate of Return (IRR) etc.

Explain



Inform the learners to identify the market price of different components of a solar PV system for a solar PV water pump.

Explain the factors Impacting the Price of Different Components of a Solar PV System in India.

Explain Key Steps to Prepare Cost Estimate for a solar PV water pumping project. Explain Cost Benefit Analysis for a Solar PV Water Pumping Project.

Notes for Facilitation



To identify the market price of different components of a solar PV system for a solar PV water pump, you can perform the following tasks:

- **Identify the required components:** Make a list of all the components required for a solar PV water pump system, such as solar panels, inverter, mounting structure, batteries, charge controller, cables, and connectors.
- **Research suppliers:** Look for reputable suppliers of solar PV components. This can be done by searching online or through industry directories. You can also contact local distributors or manufacturers of solar PV components.
- **Request quotes:** Once you have identified potential suppliers, request quotes for the specific components you need. Make sure to provide detailed specifications for each component so that you can compare quotes accurately.
- **Compare prices:** Compare the prices of the components from different suppliers. Make sure to consider factors such as quality, warranty, and delivery times when comparing prices.
- **Negotiate:** If you find that a particular supplier is offering a higher price than others, try negotiating with them to see if they can offer a lower price.
- **Make a decision:** After comparing prices and negotiating, make a decision on which supplier to purchase the components from

Demonstrate

Demonstrate the process of calculating the Levelized Cost of Energy (LCOE), Calculate the Payback Period, Calculate the Internal Rate of Return (IRR)

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.

UNIT 9.4: Select Appropriate Business Model for Solar Water Pumping

Unit Objectives

At the end of this unit, participant will be able to:

- Discuss to identify the policy, regulations and procedures for deployment of solar water pumping system.
- Explain to identify and select the appropriate business models in solar water pumping.

Explain

Inform the learners about the policy, regulations and procedures for deployment of solar water pumping system.

Explain the several business models that can be used for solar water pumping projects in India. Explain the procedure of selection of a business model for a solar PV water pump business in India

Notes for Facilitation

Selecting a Business Model for Solar Water Pumping Business

- **Access to capital:** The availability of capital will be a key factor in selecting a business model. If the farmer or project owner has access to sufficient capital, a CAPEX model may be suitable. If capital is limited, an OPEX or hybrid model may be more appropriate.
- **Financial viability:** The business model should be financially viable and should ensure that the project generates sufficient revenue to cover its costs, including the capital cost, maintenance, and operational costs. The business model should also factor in the prevailing market conditions and regulatory framework.
- **Risk and responsibility:** The business model should clearly define the roles and responsibilities of the various stakeholders, including the project owner, the third-party investor or service provider, and the farmers. The model should also factor in the risk associated with the project, including the risk of non-payment by farmers or the risk of equipment failure.
- **Long-term sustainability:** The business model should ensure the long-term sustainability of the project. It should factor in the expected lifespan of the solar PV water pump system, the availability of spare parts and maintenance services, and the future energy requirements of the farmers.
- **Social and environmental impact:** The business model should consider the social and environmental impact of the project. It should ensure that the project benefits the local community and promotes sustainable development practices.



10. Site Feasibility Study of solar PV water pump

UNIT 10.1: Assess Site for Solar PV Water Pump Installation

UNIT 10.2: Prepare Cost Benefit Analysis for the Solar Water Pump



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to assess site for solar PV pump installation.
2. Explain how to collect data of local weather conditions.
3. Explain how to conduct study for drinking and irrigation water requirement of the user/s and it is being presently met.
4. Explain how to analyse the depth of the water/ ground water source and the quantity/recharge period.
5. Explain how to assess the soil test report
6. Discuss to create relevant solar pumping model of appropriate capacity and specifications.
7. Discuss how to prepare the cost benefit analysis of the solar pump and make a presentation to the customer.
8. Discuss how to identify manufacturers and suppliers through various sources who can meet the requirements.

UNIT 10.1: Assess Site for Solar PV Water Pump Installation

Unit Objectives



At the end of this unit, the participant will be able to:

- Explain how to assess site for solar PV pump installation.
- Explain how to collect data of local weather conditions.
- Explain how to conduct study for drinking and irrigation water requirement of the user/s and it is being presently met.
- Explain how to analyse the depth of the water/ ground water source and the quantity/recharge period.
- Explain how to assess the soil test report
- Discuss to create relevant solar pumping model of appropriate capacity and specifications.

Say



Introduce the module name to the learners and the relevance of learning the basics of solar PV water pump installation.

Explain



Explain the procedure for Site Assessment for Solar PV Installation to deploy a solar-powered pumping system. Explain how to Analyse the Depth of the Water/Ground Water Source and the Quantity/Recharge Period

Inform the learner about the test are being performed while assessing the site for solar PV pump installation. Inform how to conduct a study to determine the drinking and irrigation water requirements of the user and how they are currently being met.

Notes for Facilitation



The generic formula to analyse the drinking and irrigation water requirement of a user is as follows:

- Water requirement = Number of people or animals x water consumption per person or animal x number of days of water supply

For irrigation water requirement, the formula is:

- Water requirement = Crop water requirement per day x area of land to be irrigated x number of days of irrigation

To know more about Design of Solar Pump, you may refer to the following link:

<https://www.youtube.com/watch?v=guL8Ussc5Cg>

Demonstrate

Demonstrate the procedure of Setting Solar Water Pumping Model.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the trainer and others would ask questions.

UNIT 10.2: Prepare Cost Benefit Analysis for the Solar Water Pump

Unit Objectives

At the end of this unit, the participant will be able to:

- Discuss how to prepare the cost benefit analysis of the solar pump and make a presentation to the customer.
- Discuss how to identify manufacturers and suppliers through various sources who can meet the requirements.

Explain

Inform the learners to do Cost Benefit Analysis of Solar Pump. Explain how to identify manufacturers and suppliers through various sources who can meet the requirements.

Notes for Facilitation

Estimate the savings from using a solar pump: The next step is to estimate the savings that will result from using a solar pump. This may include savings on energy costs, as well as savings on water usage.

Calculate the payback period: The payback period is the amount of time it takes for the savings from using a solar pump to offset the initial investment costs. Calculate the payback period by dividing the total costs of the solar pump by the annual savings from using the pump.

Determine the net present value: The net present value (NPV) is the difference between the present value of the benefits and the present value of the costs. Calculate the NPV by discounting the future cash flows from the savings by a discount rate to account for the time value of money.

To know more about Benefits of Solar Pumping, you may refer to the following link:

<https://www.worldbank.org/en/news/video/2016/12/19/solar-pumping>

Demonstrate

Show them how to calculate NPV, IRR. Calculate savings from using a solar pump

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.



11. Installation of Solar PV water pump

UNIT 11.1: Tools and Specifications

UNIT 11.2: Solar Pump Installation



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Discuss how to identify and use tools for solar pump installation.
2. Explain how to decide on the specification of the solar pumping system (e.g. module capacity).
3. Identify the head of the location of installation.
4. Explain how to oversee the preparation of Civil foundation for water pumping system
5. Discuss how to install the Mounting structure.
6. Ensure that the electrical installation tools are available on site
7. Discuss how to install the foundation for ACDB and Inverter
8. Discuss how to install the Earthing and Lighting arrestor on site.
9. Discuss how to install the components of the solar pumping systems; solar modules, MPPT & control electronics, inverter, pump, motor mounting structures, cables, switches etc.
10. Explain installation process of the solar water pumping system along with AC isolator and Energy meter.
11. Discuss drilling of bore of appropriate diameter and depth and fixing of pipe of appropriate diameter in the bore.
12. Discuss how to the properly place the pump into the bore

UNIT 11.1: Tools and Specifications

Unit Objectives

At the end of this unit, participant will be able to:

- Discuss how to identify and use tools for solar pump installation.
- Ensure that the electrical installation tools are available on site.
- Explain how to decide on the specification of the solar pumping system (e.g. module capacity).
- Identify the head of the location of installation.

Explain

Inform the learners about Solar Pumps. Explain components of Solar Pump. Explain different types of solar pumps. Explain working of solar pumps. Explain Block Design of Solar Water Pumping System given in slide.

Notes for Facilitation

Dynamic Head

It is calculated based on the vertical height that the water must be pumped. It is impacted by the length, material and diameter of the pipe.

Static Head

It refers to the vertical distance between the drawdown level and the highest point in the output pipe. It can be calculated using the following formula:

Static head = Drawdown level + Static water level + Lift from surface

To know more about Working of Solar PV Pump, you may refer to the following link:

https://www.youtube.com/watch?v=x_vW6YniJjY

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 11.2: Solar Pump Installation

Unit Objectives



At the end of this unit, participant will be able to:

- Explain how to oversee the preparation of Civil foundation for water pumping system
- Discuss how to install the Mounting structure.
- Discuss how to install the foundation for ACDB and Inverter
- Discuss how to install the Earthing and Lighting arrestor on site.
- Discuss how to install the components of the solar pumping systems; solar modules, MPPT & control electronics, inverter, pump, motor mounting structures, cables, switches etc.
- Explain installation process of the solar water pumping system along with AC isolator and Energy meter.
- Discuss drilling of bore of appropriate diameter and depth and fixing of pipe of appropriate diameter in the bore.
- Discuss how to the properly place the pump into the bore

Explain



Inform the learners about General Steps for Preparation of Civil Foundation for Installing Solar PV Water Pump. Explain Installation process of Solar Water Pumping System. Explain Basics of Solar Pump Selection and Unit Sizing. Explain the Procedure for Installing the Mounting Structure. Inform the learner about ACDB and DCDB Installation and Configuration. Explain Steps for connecting the junction box to the PV module.

Resources to be Used



Show them the constituents of PPE.

Tell them to wear PPE only when required and replace the damaged one.

Ask



You may question the learners if they have any idea about the role of solar PV Water Pump installer in terms of the activities needed to be performed

Notes for Facilitation

Drilling of Bore and Placing Pump into the Bore

Drilling a borehole and fixing a pipe of appropriate diameter in it is an important step in the installation of a water pump system. Here are some key considerations when drilling a borehole and fixing a pipe:

- Determine the appropriate diameter and depth for the borehole: The diameter and depth of the borehole will depend on factors such as the water demand, the soil type, and the water table level. It's important to consult with a qualified drilling contractor to determine the appropriate dimensions for the borehole.
- Drill the borehole: Once the dimensions are determined, the borehole can be drilled using a drilling rig. The drilling process can take several hours or even days, depending on the depth and the geology of the soil.
- Clean the borehole: After drilling, the borehole should be cleaned to remove any debris and ensure a clear path for water to flow. This can be done using a high-pressure water jet or an air compressor.
- Install the casing pipe: A casing pipe of appropriate diameter should be inserted into the borehole and anchored to prevent it from collapsing or shifting. The casing pipe should be of high-quality material to prevent corrosion and ensure a long lifespan.
- Install the screen pipe: A screen pipe should be installed inside the casing pipe to allow water to flow into the pump without allowing sand, silt or other debris to enter. The screen pipe should be sized appropriately to match the pumping capacity of the system.
- Grout the borehole: Grouting is the process of filling the space between the casing pipe and the borehole wall with a special cement mixture. This helps to prevent surface water contamination and stabilize the casing pipe.
- Test the borehole: Once the borehole is drilled, the casing pipe and screen pipe are installed, and the borehole is grouted, it's important to test the borehole to ensure that it is producing the desired amount of water. A pump test can be performed to determine the flow rate, pumping capacity, and drawdown of the borehole

To know more about Install Components of Solar Pumping Systems, you may refer to the following link:

<https://www.youtube.com/watch?v=DbJk1M4U7W4>

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.



12.Operation and maintenance of Solar

UNIT 12.1: Check Integrity and Working Condition of all Components

UNIT 12.2: Troubleshooting the Identified Faults

UNIT 12.3: Maintenance Procedure



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to check the integrity and working condition of all components, fuses and circuit breakers
2. Ensure the working condition of the junction boxes, MPPT, controller and all other key electrical components
3. Ensure the continuity of cables and wires are maintained.
4. Ensure proper electrical connections throughout the solar PV pump to Inverter and controller
5. Explain how to troubleshoot the identified faults in controller and PV PUMP
6. Discuss how to ensure proper cleaning of modules as per schedule and perform standard procedure
7. Explain how to record key system parameters and perform preventive and predictive maintenance procedures

UNIT 12.1: Check Integrity and Working Condition of all Components

Unit Objectives



At the end of this unit, participant will be able to:

- Explain how to check the integrity and working condition of all components, fuses and circuit breakers
- Ensure the working condition of the junction boxes, MPPT, controller and all other key electrical components
- Ensure the continuity of cables and wires are maintained.
- Ensure proper electrical connections throughout the solar PV pump to Inverter and controller

Say



Introduce the module name to the learners and ensure the relevance of working condition of the junction boxes, MPPT, controller and all other key electrical components.

Demonstrate



Demonstrate the thermography procedure of the PV system.

To know more about Anti-Islanding Protection, you may refer to the following link:

<https://www.youtube.com/watch?v=Ulot5MUJFbY>.

Explain



Explain thermography and Explain benefits of thermography. Explain how to check the working condition of Junction Boxes, MPPT, Controller.

Inform the learner how to Check the Electrical Connections Through Solar PV to Inverter and Controller

Ask



You may question the learners if they have any idea about the electrical connections in terms of the activities needed to be performed.

Notes for Facilitation

Maintenance of Cables, Connectors And Switches

- Visually check all conduit and wire insulation for damage.
- Check for loose, broken, corroded, or burnt wiring connections.
- Check if all equipments are connected with proper wire and conduit.
- Make sure all wiring is secured, by gently but firmly pulling on all connections.
- Check all terminals and wires for loose, broken, corroded, or burnt connections or components.
- Check the operation of the PV array DC isolators
- Check array wiring for physical damage and wind chafing

Check the working condition of Circuit Breakers

To check the working of circuit breaker, you need to perform the following tasks:

- Open the circuit breaker box, unscrew the metal frame around the circuit breakers to access the wires attached through the circuit breakers.
- To check the voltage, set the Multimeter to Volts AC.
- Attach the prongs to each side of the circuit breaker. If the reading are between 120 to 140 Volts than the circuit breaker is working perfectly. But, any reading beyond this indicates a faulty circuit breaker.
- To check the resistance, set the Multimeter to Ohm.
- Attach the prongs to each side of the circuit breaker. If the reading is zero (0) Ohm that indicates the circuit breaker is in good condition.
- But, if the reading is 0.01 ohms, this indicates too much resistance within the breaker and this may be the issue that needs to be addressed immediately.

Check the Working Condition of Controller

- Testing procedure for shunt charge controllers (12V system):
- **STEP 1:** Set multimeter to appropriate DC voltage range to measure the voltage between the array positive and array negative (terminals).
- **STEP 2:** Measure the DC voltage between the battery positive and battery negative terminals on the controller. If the controller is operating properly, it should be between 13.5 and 14.5 volts per module in series.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 12.2: Troubleshooting the Identified Faults

Unit Objectives



At the end of this unit, participant will be able to:

- Explain how to troubleshoot the identified faults in controller and PV PUMP

Explain



Tell the learners about the troubleshoot mechanism for the identified faults in PV PUMP.
Explain Troubleshoot Mechanism for the Identified Faults in Controller.

Demonstrate



Demonstrate the process of troubleshooting in pump and controller.

Say



Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Role Play



Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the Trainer and others would ask questions.

UNIT 12.3: Maintenance Procedure

Unit Objectives

At the end of this unit, participant will be able to:

- Discuss how to ensure proper cleaning of modules as per schedule and perform standard procedure
- Explain how to record key system parameters and perform preventive and predictive maintenance procedures

Explain

Tell the learners about the maintenance requirement of different components.

Explain about the maintenance checklist of charge controllers, batteries, Solar PV module, wires.

Explain the procedure of preparing the maintenance checklist.

Demonstrate

Demonstrate the process of troubleshooting in pump and controller.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the Trainer and others would ask questions.



13. Solar PV Water Pump Project lifecycle

UNIT 13.1: Read and Interpret the Diagram, Drawings and Bill of Material

UNIT 13.2: Lifecycle of a Solar PV Water Pump Project



Key Learning Outcomes



At the end of this module, the participant will be able to:

1. Discuss how to read and interpret the single line diagram, civil / mechanical drawings and electrical drawings.
2. Explain how to read and interpret the bill of material.
3. Explain how to calculate the lifecycle cost of a solar water pump.
4. Identify and mitigate various risks associated with the project.
5. Discuss how to ensure the solar water pump and structure meets the local government and regulatory requirements.
6. Prepare action plan and coordinate the implementation of solar water pumping project.
7. Discuss how to identify the maintenance activity required for a solar water pump component.
8. Prepare a preventive maintenance schedule.
9. Ensure proper cleaning of solar panels periodically.
10. Ensure regular inspection of the solar water pump to identify and rectify the faults.

UNIT 13.1: Read and Interpret the Diagram, Drawings and Bill of Material

Unit Objectives



At the end of this unit, participant will be able to:

1. Discuss how to read and interpret the single line diagram, civil / mechanical drawings and electrical drawings.
2. Explain how to read and interpret the bill of material.

Explain



Tell the learners about the Procedure to read diagrams, drawings, and bills of materials (BOMs) in a Solar PV (photovoltaic) pump system.

Explain Water Pump Controller Circuit using Solar Panel as given in module.

Demonstrate



Demonstrate the drawing of water pump controller circuit using solar panel. Demonstrate sample bill of material

Say



Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

Role Play



Educate them on how to ask the previously explained pertinent questions.
Then do a role play, where one learner would be the Trainer and others would ask questions.

UNIT 13.2: Lifecycle of a Solar PV Water Pump Project

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain how to calculate the lifecycle cost of a solar water pump.
2. Identify and mitigate various risks associated with the project.
3. Discuss how to ensure the solar water pump and structure meets the local government and regulatory requirements.
4. Prepare action plan and coordinate the implementation of solar water pumping project.

Explain

Tell the learners about the methods for evaluating Costing of Solar PV Water Pump Project.

Explain Business Models for Solar PV Water Pump System. Explain the type of Associate with the Project and the methods to mitigate risks.

Notes for Facilitation

Costing of Solar PV Water Pump Project includes:

- Life Cycle Cost Analysis (LCCA) - This is the total of all expenses and benefits related to the pumping system over the course of its life or over a chosen period of analysis, expressed in present-day money. This is called Net Present Value (NPV)
- Pay back term - The amount of time needed for the initial investment to be fully repaid by the savings or benefits received.
- Total cost savings: It refers to the difference between the total costs of two investments (resulting from the adoption of two different technologies) at the conclusion of the assessment period.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.



14. Site Feasibility Study for Ground Mount Solar power plant

UNIT 14.1: Site Feasibility Study for Ground Mount Solar power plant

UNIT 14.2: Soil Testing for Ground Mount Solar power plant.



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to assess irradiation and other geological data
2. Explain how to assess the daily, monthly and annual solar resource data including GHI, DNI, Albedo etc. for site to evaluate the potential for solar energy generation at the site.
3. Explain how to collect data of local weather conditions such as temperature range, flooding, wind speed, humidity, pollution levels, snow and other climatic conditions for assessment of its impact on solar energy generation
4. Explain how to conduct the soil testing and contour mapping.
5. Explain how to conduct the detailed survey plan of the land proposed for installation of solar power plant with
6. elevations and topography contour mapping etc.
7. Discuss how to assess physical site accessibility
8. Discuss to identify the policy, regulations and procedures for ground mounted solar sector.
9. Discuss how to assess transmission line & power evacuation

UNIT 14.1: Site Feasibility

Unit Objectives



At the end of this unit, participant will be able to:

1. Explain how to assess irradiation and other geological data
2. Explain how to assess the daily, monthly and annual solar resource data including GHI, DNI, Albedo etc. for site to evaluate the potential for solar energy generation at the site.
3. Explain how to collect data of local weather conditions such as temperature range, flooding, wind speed, humidity, pollution levels, snow and other climatic conditions for assessment of its impact on solar energy generation

Explain



Explain Solar Irradiance and different components of Solar Irradiance. Explain Albedo and general steps to measure albedo.

Tell the Learner about the methods that can be used to collect data on local weather conditions.

Ask



You may question the learners if they have any idea about the role of solar irradiation and geological data in terms of the activities needed to be performed.

Notes for Facilitation



- **DNI:** It stands for Diffused Normal Irradiance. It represents the amount of sunlight coming perpendicular to a surface. DNI refers to sunrays that come in a straight line from the direction of the sun at its current position in the sky. The Solar PV panels maximize DNI by means of tilting or rotating with the angle of sun.
- **DHI:** It stands for Diffused Horizontal Irradiance. It represents solar radiation that does not arrive on a direct path from the sun, but is scattered by clouds or other particles in the atmosphere and comes equally from all directions.
- **GHI:** It stands for Global Horizontal Irradiance. It represents the total amount of shortwave radiation received from above by a surface which is horizontal to the surface.
As per mathematical equation:
 - **Global Horizontal Irradiance (GHI) = Direct Normal Irradiance (DNI)* cos (solar zenith angle) + Diffused Horizontal Irradiance (DHI)**

UNIT 14.2: Soil Testing

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to conduct the soil testing and contour mapping.
- Explain how to conduct the detailed survey plan of the land proposed for installation of solar power plant.

Ask

You may question the learners if they have any idea about the role of Soil Testing in terms of the activities needed to be performed.

Explain

Explain Common methods for soil testing. Explain Standard Penetration test, cone penetration test. Describe the image given on slide of standard penetration test. Explain Topography, Contour Mapping and Its Importance in Solar Plant Design. Explain Importance of Topography in Solar Design

Notes for Facilitation

The following list of factors illustrates the significance of contour mapping in solar plant design:

- **Site Selection:** Contour mapping can help in locating the solar plant's ideal location. The contour lines can be used by solar plant designers to assess a site's suitability as a location for the facility and to identify any potential difficulties that might arise during construction.
- **Maximum Solar Energy Yield and Optimal Solar Panel Orientation:** Contour maps assist in locating the regions on the site where the solar panels can be positioned for maximum solar energy yield and optimum orientation. The designers can choose the best tilt and orientation for the solar panels to produce the most solar energy by using the contour lines, which display the differences in elevation across the site.
- **Drainage and erosion control:** Contour mapping helps in locating regions with a steep incline and the site's natural water flow. Designing efficient drainage systems and erosion control techniques is essential for reducing soil erosion and sedimentation.
- **Accurate Cost Estimation:** Contour mapping facilitates the development of accurate estimates of costs for the solar plant's construction. Designers can precisely predict the amount of earthwork needed, the price of retaining walls, and other building expenditures thanks to the contour map's extensive information on the topography of the site.

Demonstrate

Demonstrate Standard Penetration Test and Cone Penetration Test (CPT).

To know more about Standard Penetration Test, you may refer to the following link:

<https://www.youtube.com/watch?v=DjWDOqQjsyQ>

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.



15. Interpret and prepare Civil foundation, layout, mounting structure and allied civil work in ground mount solar plant

UNIT 15.1: How to Prepare a Project Development Plan.

UNIT 15.2: Mounting Structure Design and Soil Test Report

UNIT 15.3: Interpret Layout, Mounting Structure and Civil Work Design



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to prepare a project development plan.
2. Discuss how to assess soil characteristics in accordance to mounting structure design requirement.
3. Discuss the type of foundation suitable for module mounting structures, inverters, transformers etc. based on the soil test report.
4. Discuss how to prepare structural design report.
5. Discuss how to prepare the layout for cable trenching, Electrical control panel, Transformer, VCB etc.
6. Explain how to prepare Pile foundation, column post and other civil foundation.
7. Explain the design of the pile foundation as per the soil assessment report.
8. Explain how to assess design of the civil/ structural allied works of the solar PV power plant including compound wall/entry gate internal plant roads walkways between different rows of modules water distribution network water drainage system, etc.
9. Explain the importance of soil test report for designing the Mounting structure.
10. Discuss civil/ structural drawings for the ground mount solar power plant.
11. Discuss the reinforcement material for civil construction of various foundation.

UNIT 15.1: How to Prepare a Project Development Plan?

Unit Objectives



At the end of this unit, participant will be able to:

- Explain how to prepare a project development plan.

Explain



Explain Steps to Create Project Development Plan. Explain how to identify Project Resources.

Explain the steps given on the slide. Explain how to monitor and evaluate Project Progress.

Ask



You may question the learners if they have any idea about Project development plan in terms of the activities needed to be performed.

Role Play



Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the trainer and others would ask questions.

Summarize



Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 15.2: Mounting Structure Design and Soil Test Report

Unit Objectives



At the end of this unit, participant will be able to:

- Discuss how to assess soil characteristics in accordance to mounting structure design requirement.
- Explain the importance of soil test report for designing the Mounting structure.
- Discuss the type of foundation suitable for module mounting structures, inverters, transformers etc. based on the soil test report.
- Discuss how to prepare structural design report.
- Discuss how to prepare the layout for cable trenching, Electrical control panel, Transformer, VCB etc.
- Explain how to prepare Pile foundation, column post and other civil foundation.
- Explain the design of the pile foundation as per the soil assessment report.

Explain



Explain how to assess soil characteristics. Explain Soil Composition by Volume. Explain different types of soil structure. Explain Soil Bearing Capacity. Explain different types of foundations. Explain how to prepare structural design report. Explain the Procedure for Preparing the Layout for Cable Trenching in solar PV Plant.

Demonstrate



Demonstrate the Procedure for Layout for Cable Trenching in solar PV Plant. Demonstrate the Procedure for Preparing Pile Foundation, Column Post and Other Civil Foundation

Do



You may show a small video on the process of Civil Foundations for Ground Mounted Solar Plant using the following link:

<https://www.youtube.com/watch?v=miciDafdy1w>

Tell them about the formula of converting temperature in Kelvin to temperature in degree Celsius i.e. $K = 273 + \text{temperature in degree Celsius}$

Notes for Facilitation

Start by giving an introduction about yourself to the learners, using the same points given on the slide.

Preparing Pile Foundation

Preparing a pile foundation for a ground-mounted solar plant installation involves several steps:

- **Site Evaluation:** Conduct a site survey to assess the terrain and soil characteristics. This survey should include soil testing to determine the soil's bearing capacity, depth to bedrock, and other geotechnical parameters.
- **Pile Selection:** Select the type of pile to be used based on the site's soil conditions, foundation loads, and other factors. Common pile types used for solar plant installations include steel H-piles, concrete piles, and screw piles.
- **Pile Design:** Design the pile foundation based on the site's soil characteristics, size of the solar panels, and other factors. The foundation should be designed to support the loads imposed by the solar panels and distribute them evenly over the soil.
- **Pile Installation:** Install the piles using the appropriate equipment, such as a pile driver or drilling rig. The piles should be driven or drilled to the required depth to ensure adequate support.
- **Pile Cap:** Install a pile cap to distribute the loads from the solar panel mounting structure evenly across the pile foundation.
- **Mounting Structure:** Install the solar panel mounting structure on the pile foundation using anchor bolts or other appropriate fasteners.
- **Electrical Grounding:** Ground the mounting structure and solar panels to ensure electrical safety.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 15.3: Interpret Layout, Mounting Structure and Civil Work Design

Unit Objectives



At the end of this unit, participant will be able to:

1. Explain how to assess design of the civil/ structural allied works of the solar PV power plant including compound wall/entry gate internal plant roads walkways between different rows of modules water distribution network water drainage system, etc.
2. Discuss civil/ structural drawings for the ground mount solar power plant.
3. Discuss the reinforcement material for civil construction of various foundation.

Explain



Explain guidelines for Assessment Criteria for Assessing Compound Wall/Entry Gate. Explain Assessment Criteria for Internal Roads of Solar Plant. Explain Assessment Criteria for Assessing the Walkways Between Different Rows of Modules. Explain Assessment Criteria for Assessing the Water Drainage System. Explain about Reinforcement Material for Civil Construction of Various Foundation.

Ask



You may question the learners if they have any idea about the role of civil/ structural allied works in terms of the activities needed to be performed.

Resources to be Used



Show them the constituents of PPE.

Tell them to wear PPE only when required and replace the damaged one.

Notes for Facilitation

The type of foundation and the loads it will support will determine the choice of reinforcement material. Following are some examples:

- **Pile Foundation:** Steel bars are frequently utilised as reinforcement material for piling foundations. Prior to pouring the concrete, these steel bars are inserted into the piling borehole. The steel reinforcement aids in the pile's ability to withstand tensile forces over the course of its lifetime. The size of the pile and the loads it will support will determine the thickness and quantity of steel bars to be employed.
- **Strip Foundation:** Prior to pouring the concrete in the foundation trenches for strip foundations, steel reinforcing is inserted. The reinforcement aids in resisting the tensile forces that earth movement or settlement will subject the foundation to. The foundation's dimensions and the loads it will support determine the steel reinforcement's thickness and spacing.
- **Raft Foundation:** Steel mesh or steel bars are frequently utilised as reinforcement material for raft foundations. To enhance strength and avoid cracking, the reinforcement is inserted inside the concrete raft. The raft size and the loads it will support determine the thickness and spacing of the steel reinforcement.
- **Retaining Wall:** Steel bars are inserted horizontally into retaining walls to act as reinforcement. The steel reinforcement aids in resisting the tensile pressures that soil pressure will cause to act on the retaining wall. The wall height and the loads it will support determine the thickness and spacing of the steel reinforcement.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.



SGJ/N0403

Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to prepare a project development plan
2. Discuss the planning for construction of compounds wall or fencing, access of internal road, drainage
3. Explain how to develop the overall project constructability approach for cost effective implementation
4. Explain how to supervise and manage schedule for installation of module structures and modules, installation of inverters, transformers, earthing piles, DC/ AC power protection devices, lightning arresters, substation as per the grid codes and regulatory provisions
5. Discuss how to plan, schedule, and monitor all material deliveries to avoid project delays
6. Discuss how to manage schedule for each of the construction activity i.e. construction of internal roads, construction of foundations for mounting module support structures, combiner boxes, inverters, transformers, and substation etc.
7. Discuss how to do successful completion of the projects by identifying innovative ways to maximize Profit
8. Explain how to manage project billing cycle to ensure timely submission and payment of invoices
9. Explain how to analyze the bids to ensure complete and accurate quotes for scopes of work
10. List various parameters associated with local power loads/ grid dispatch
11. Explain how to negotiate with subcontractors
12. Discuss how to coordinate with the customers on a regular basis and provide them with the updates
13. Discuss how to communicate to the customer and other key team members the contractual, financial, cost accounting, scheduling, technical and other matters related to the project.
14. Discuss periodic visits to the site(s) to review the progress at the ground

UNIT 16.1: Planning and Scheduling

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to prepare a project development plan
- Discuss the planning for construction of compounds wall or fencing, access of internal road, drainage
- Explain how to develop the overall project constructability approach for cost effective implementation
- Discuss how to plan, schedule, and monitor all material deliveries to avoid project delays

Ask

You may question the learners if they have any idea about the role of Project constructability and cost-effective implementation in terms of the activities needed to be performed.

Explain

Explain how to prepare a Project Development Plan for Ground-Mounted Solar Civil Work Construction. Explain how to develop the overall project constructability approach for cost effective implementation. Explain how to plan, schedule, and monitor all material deliveries to avoid project delays.

Resources to be Used

Show them the constituents of PPE.

Tell them to wear PPE only when required and replace the damaged one.

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.

UNIT 16.2: Managing Ground Mounted Solar Civil Construction Work

Unit Objectives



At the end of this unit, participant will be able to:

- Explain how to supervise and manage schedule for installation of module structures and modules, installation of inverters, transformers, earthing piles, DC/ AC power protection devices, lightning arresters, substation as per the grid codes and regulatory provisions
- Discuss how to manage schedule for each of the construction activity i.e. construction of internal roads, construction of foundations for mounting module support structures, combiner boxes, inverters, transformers, and substation etc.
- Discuss how to do successful completion of the projects by identifying innovative ways to maximize Profit
- Explain how to manage project billing cycle to ensure timely submission and payment of invoices
- Explain how to analyze the bids to ensure complete and accurate quotes for scopes of work
- Explain how to negotiate with subcontractors
- List various parameters associated with local power loads/ grid dispatch
- Discuss how to coordinate with the customers on a regular basis and provide them with the updates
- Discuss how to communicate to the customer and other key team members the contractual, financial, cost accounting, scheduling, technical and other matters related to the project.

Explain



Explain how to supervise and manage schedule for installation of module structures and modules, installation of inverters, transformers, earthing piles, DC/ AC power protection devices, lightning arresters, substation as per the grid codes and regulatory provisions.

Tell the learner about Grid Codes and Regulatory Provisions for the Installation of Components.

Explain key steps to identify innovative ways to maximize profit during project execution. Explain how to analyse bids. Explain how to negotiate with subcontractors

Ask



You may question the learners if they have any idea about the Grid Codes and Regulatory Provisions role of solar PV plant in terms of the activities needed to be performed.

Notes for Facilitation

Steps to Analyse Bids

- Determine project requirements: The entrepreneur should be clear about the requirements of the project, including the scope of work, timelines, and budget.
- Identify potential vendors or contractors: Entrepreneurs should research and identify potential vendors or contractors who can meet the requirements of the project.
- Invite bids: Once potential vendors or contractors have been identified, entrepreneurs should invite them to submit bids. The bids should clearly specify the scope of work, timelines, and cost.
- Evaluate the bids: Entrepreneurs should review and evaluate the bids received from the vendors or contractors. The evaluation should consider factors such as the quality of work, experience, and reputation of the vendor or contractor, the proposed timeline, and the cost.
- Compare the bids: Entrepreneurs should compare the bids received from different vendors or contractors to identify any differences in their approach, pricing, and overall quality. This can help to determine which vendor or contractor is best suited for the project.
- Negotiate and finalize the contract: Once a vendor or contractor has been selected, entrepreneurs should negotiate the terms of the contract to ensure that the scope of work, time frame, and budget are clearly defined. This can help to prevent misunderstandings and disputes later on in the project.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.



17. Electrical Feasibility Study for Ground Mount Solar Power Plant

UNIT 17.1: Electrical Feasibility Report for Ground Mount Solar Power Plant

UNIT 17.2: Estimate Energy Generation from the Solar PV Power Plant.



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to assess site electrical feasibility
2. Explain how to assess cable root
3. Discuss how to identify inverter location
4. Discuss how to identify the transformer and other HT equipment location
5. Explain how to assess grid availability for power evacuation including nearest substation and transmission line capacity as well as distance from project site

Explain how to assess the evacuation voltage level i.e. 230 V/440 V/11 KV /33 KV66 kV

UNIT 17.1: Electrical Feasibility Report for Ground Mount Solar Power Plant

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to assess site electrical feasibility
- Explain how to assess cable route
- Discuss how to identify inverter location
- Discuss how to identify the transformer and other HT equipment location

Ask

You may question the learners if they have any idea about the Electrical Feasibility Report for Ground Mount Solar Power Plant in terms of the activities needed to be performed.

Explain

Explain Electrical Feasibility Report for Ground Mount Solar Power Plant. Tell the learners about Importance of Electrical Feasibility Report.

Tell the learners about the Parameters Included in an Electrical Feasibility Report for Ground Mount Solar Power Plant.

Explain the steps to conduct a site electrical feasibility assessment. Explain the steps to assess the cable route

Notes for Facilitation

- **Grid Availability Assessment:**
 - Identify the nearest substation and assess its capacity to accommodate the additional power generated by the solar power plant.
 - Evaluate the distance between the project site and the substation, considering transmission line capacity and potential grid constraints.
 - Verify the availability of grid connection points and the feasibility of connecting the solar power plant to the grid.

To know more about cable route for a ground mount solar power plant, you may refer to the following link:

<https://www.youtube.com/watch?v=kazwzLyKD2A>

UNIT 17.2: Estimate Energy Generation from the Solar PV Power Plant

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to assess grid availability for power evacuation including nearest substation and transmission line capacity as well as distance from project site
- Explain how to assess the evacuation voltage level i.e. 230 V/440 V/11 KV /33 KV66 kV

Explain

Explain the tasks involved in estimating energy generation. Explain the various steps involved in calculating the energy generation from the Solar PV Plant. Explain the procedure to calculate radiation on panels.

Tell the learners about the Evacuation Voltage Level. Explain various factors on which evacuation voltage level depends.

Ask

You may question the learners if they have any idea about the role of Power evacuation in terms of the activities needed to be performed.

Notes for Facilitation

Solar radiation (kWh/m²/day) = $I_{sc} \times (1 + a \times \text{Tilt}) \times (1 - b \times |\text{Lat} - \text{Lat0}|)$

Where:

- o I_{sc} = Extraterrestrial radiation on a horizontal surface (typically available from solar radiation databases)
- o Tilt = Tilt angle of the solar panel (in degrees)
- o a, b = Empirical coefficients for the latitude adjustment factor
- o Lat = Latitude of the project location
- o Lat0 = Latitude of the reference location

Radiation on panels (kWh/m²/day) = Solar radiation (kWh/m²/day) $\times \cos(\text{Incidence Angle})$

Where:

- Incidence Angle = Angle between the solar radiation vector and the normal vector of the panel surface



18. Interpret drawings of DC and AC side of electrical Component of Ground Mount Solar Power Plant

UNIT 18.1: Interpret Drawings of DC and AC Side of Electrical Components of Ground- Mount Solar Power Plant

UNIT 18.2: Electrical installation, BOQ, Commissioning of Solar Plant



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to interpret the electrical single line diagram of DC side of the interconnection
2. Interpret the electrical line diagram of ACDB, LT Panel, earthing layout.
3. Discuss how to install and interconnect the Inverter, DC-Disconnect, String monitoring box (SMB)
4. Discuss how to install the DC and AC Energy Meter
5. Discuss how to ensure the Quality and workmanship of Cable laying and other electrical installation
6. Interpretate the electrical single diagram of AC side of the plant.
7. Explain how to prepare BOQ with the with as built drawing of electrical interconnection and internal wiring diagram of major electrical component
8. Discuss the process of Pre commissioning and commissioning activities of solar Plant in both DC and AC side.

UNIT 18.1: Interpret Drawings of DC and AC Side of Electrical Components of Ground- Mount Solar Power Plant

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain how to interpret the electrical single line diagram of DC side of the interconnection
2. Interpret the electrical line diagram of ACDB, LT Panel, earthing layout.
3. Discuss how to install and interconnect the Inverter, DC-Disconnect, String monitoring box (SMB)
4. Discuss how to install the DC and AC Energy Meter
5. Interpretate the electrical single diagram of AC side of the plant.
6. Explain how to prepare BOQ with the with as built drawing of electrical interconnection and internal wiring diagram of major electrical component
7. Discuss the process of Pre commissioning and commissioning activities of solar Plant in both DC and AC side.

Ask

You may question the learners if they have any idea about the role of Interpretation of SLD and interconnections in terms of the activities needed to be performed.

Explain

Tell the learner about the key components and their functions in generating and transmitting electricity.

Explain the electrical distribution and safety systems within a ground-mounted solar power plant.

Resources to be Used

Show them the constituents of PPE.

Tell them to wear PPE only when required and replace the damaged one

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 18.2: Electrical installation, BOQ, Commissioning of Solar Plant

Unit Objectives

At the end of this unit, participant will be able to:

1. Discuss how to ensure the Quality and workmanship of Cable laying and other electrical installation
2. Explain how to prepare BOQ with the with as built drawing of electrical interconnection and internal wiring diagram of major electrical component
3. Discuss the process of Pre commissioning and commissioning activities of solar Plant in both DC and AC side.

Explain

Explain key considerations to ensure high-quality installations. Tell the learner about the steps to Prepare BOQ as per built drawing of Electrical Interconnection.

Explain the pre-commissioning and commissioning activities of a solar power plant

Ask

You may question the learners if they have any idea about the role of solar PV installer in terms of the activities needed to be performed.

Resources to be Used

Show them the constituents of PPE.

Tell them to wear PPE only when required and replace the damaged one.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.



19. Manage Electrical installation in Ground Mount Solar Plant

UNIT 19.1: Prepare a vendor list of potential suppliers and negotiate with them

UNIT 19.2: Manage Electrical installation in Ground mount solar plant



Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Discuss how to identify manufacturers and suppliers through various sources who can meet the requirements.
2. Ensure the electrical installation tools shall be available on site.
3. Ensure the knowhow of all electrical tools utilised for system installation.
4. Explain the testing and commissioning of equipment in DC and AC side of the plant.
5. Explain how to commissioned the transformer and other HT equipment.
6. Ensure the electrical safety at site
7. Discuss how to get the CEIG/Electrical inspector inspection done at site.

UNIT 19.1: Prepare a vendor list of potential suppliers and negotiate with them

Unit Objectives

At the end of this unit, participant will be able to:

1. Discuss how to identify manufacturers and suppliers through various sources who can meet the requirements.
2. Discuss how to manage vendor or supplier for material procurement.

Ask

You may question the learners if they have any idea about preparing the vendor or supplier for material procurement in terms of the activities needed to be performed.

Explain

Tell the learners about the steps to identify manufacturers and suppliers through various sources. Explain how to effectively manage vendors or suppliers.

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 19.2: Manage Electrical installation in Ground mount solar plant

Unit Objectives

At the end of this unit, participant will be able to:

- Ensure the knowhow of all electrical tools utilised for system installation.
- Explain the testing and commissioning of equipment in DC and AC side of the plant.
- Explain how to commissioned the transformer and other HT equipment.
- Ensure the electrical safety at site
- Discuss how to get the CEIG/Electrical inspector inspection done at site.

Ask

You may question the learners if they have any idea about the role of managing Electrical installation in Ground mount solar plant in terms of the activities needed to be performed.

Explain

Tell the learners about Know-how of all electrical tools utilised for Solar Ground mount system installation. Explain the procedure for Testing and commissioning of equipment in DC and AC side of the plant. Explain the general steps to commissioned the transformer and other HT equipment.

Tell the learner about the general steps to ensure compliance with regulations and facilitate a smooth inspection process.

Resources to be Used

Show them the constituents of PPE.

Tell them to wear PPE only when required and replace the damaged one.

Notes for Facilitation

DC Side Testing and Commissioning:

- a. **Inspection of Solar Panels:** Verify that all solar panels are correctly installed, free from damage, and oriented optimally for sunlight exposure.
- b. **String Testing:** Conduct continuity tests to ensure proper wiring connections within each string of solar panels. Use a multimeter to check voltage and current levels.
- c. **DC Isolation Testing:** Test the isolation of the DC side components (panels, strings, and inverters) from the ground and ensure there are no leakages using an insulation resistance tester.
- d. **Inverter Testing:** Verify the correct installation and functionality of inverters. Test for output voltage, frequency, and ground fault protection.
- e. **Array Grounding System Testing:** Ensure the effectiveness of the grounding system to mitigate the risk of electrical hazards.
- f. **DC System Performance Check:** Measure the actual output of the DC side under standard conditions to ensure it matches expected values based on design.

AC Side Testing and Commissioning:

- a. **Inverter Grid Connection Testing:** Verify the inverter's connection to the grid and check synchronization with the grid frequency and voltage levels.
- b. **AC Isolation Testing:** Ensure the AC side components (inverters, transformers, switchgear) are isolated from the ground and perform insulation resistance tests.
- c. **Protection System Testing:** Test protective devices like circuit breakers, relays, and fuses to ensure they operate correctly in case of overloads or faults.
- d. **Metering and Monitoring Systems:** Validate the accuracy of energy meters and monitoring systems to track power generation accurately.
- e. **Grid Interconnection Testing:** Coordinate with utility companies to conduct tests and obtain necessary approvals for grid interconnection.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.





Employability Skills

Kindly refer to the Facilitator guide of Employability Skills
available on

[Employability skills workbook](#)







Annexure



Module No.	Unit No.	Topic Name	URL	QR Code(s)
Module 1 - Introduction to Solar PV Sector in India	Unit 1.1	Types of Solar PV Systems	https://www.youtube.com/watch?v=FurOeegcA3Q	
Module 1 - Introduction to Solar PV Sector in India	Unit 1.1	Types of Solar Panel	https://www.youtube.com/watch?v=qlpRaQkzqAw	
Module 1 - Introduction to Solar PV Sector in India	Unit 1.2	Business Matters Where does India stand on solar energy targets	https://www.youtube.com/watch?v=wAxxyl-3_uc	
Module 1 - Introduction to Solar PV Sector in India	Unit 1.2	Introduction to the Renewable Energy Financial Model	https://www.youtube.com/watch?v=n0MF0iZEOes	
Module 1 - Introduction to Solar PV Sector in India	Unit 1.2	Solar Energy vs. Fossil Fuels: A Comparison	https://www.youtube.com/watch?v=u-cq7TwK3Xk	
Module 1 - Introduction to Solar PV Sector in India	Unit 1.3	Understanding Grid Interactive Systems	https://www.youtube.com/watch?v=zr9AlfA85ME	
Module 2 -	Unit 2.6	Keys to Solar Marketing and Business Success	https://www.youtube.com/watch?v=W3QuS47UrpQ	

Module 3 - Maintain Personal Health & Safety at Project Site	Unit 3.4	Housekeeping Training Video	https://www.youtube.com/watch?v=EU0ZcSQPwE4	
Module 4 - Market Research and Cost Estimation of Rooftop Solar Project	Unit 4.3	Cost benefit analysis of Rooftop PV system	https://www.youtube.com/watch?v=cTJNPOa1hrl	
Module 5 -	Unit 5.1	Mounting Solar Panels	https://www.youtube.com/watch?v=QujOA2TJm3k	
Module 5 -	Unit 5.1	How to read and understand Electricity Bill	https://www.youtube.com/watch?v=qiBvhNKd3JY	
Module 6 - Installation of Solar PV rooftop plant	Unit 6.2	Detailed Solar Inverter Installation: A Step-by-Step Guide	https://www.youtube.com/watch?v=2D8bMRZ280I	
Module 6 - Installation of Solar PV rooftop plant	Unit 6.2	Permanent Roof Anchor Installation	https://www.youtube.com/watch?v=nw-ZM7hLfxU	
Module 6 - Installation of Solar PV rooftop plant	Unit 6.3	Solar Plant Earthing	https://www.youtube.com/watch?v=GQMgMZn3WBU	
Module 6 - Installation of Solar PV rooftop plant	Unit 6.4	How to test a solar panel.	https://www.youtube.com/watch?v=eh6hzMb69gY	

Module 7 - Operation and Maintenance of Solar PV Rooftop Plant	Unit 7.2	Testing of cable/ wire	https://www.youtube.com/ watch?v=cgqzezYaYkc	
Module 7 - Operation and Maintenance of Solar PV Rooftop Plant	Unit 7.3	Equipment Grounding for PV Systems	https://www.youtube.com/ watch?v=-kjM9VWKHnk	
Module 7 - Operation and Maintenance of Solar PV Rooftop Plant	Unit 7.3	Solar Operations and Maintenance	https://www.youtube.com/ watch?v=pmWlvUWkeQo	
Module 8 - Manage Solar PV Project Lifecycle	Unit 8.1	Solar module BOM	https://www.youtube.com/ watch?v=TraDKXKvPa8	
Module 8 - Manage Solar PV Project Lifecycle	Unit 8.3	Rooftop Solar PV Systems	https://www.youtube.com/ watch?v=gX4MmPkaWDk	
Module 8 - Manage Solar PV Project Lifecycle	Unit 8.3	How to Troubleshoot Solar PV System Assets	https://www.youtube.com/ watch?v=MLNVA_iusAs	
Module 9 - Market Research and Cost Estimation of Solar PV Water Pump Business	Unit 9.1	PM KUSUM Scheme Explained in detail with SWOT analysis	https://www.youtube.com/ watch?v=ujk5cRoei9I	

Module 9 - Site Feasibility Study of solar PV water pump	Unit 9.4	Solar Water Pumping Life Cycle Cost Analysis comparisons	https://www.youtube.com/watch?v=q2vYCTEtTdg	
Module 10 - Site Feasibility Study of solar PV water pump	Unit 10.1	Design of Solar Pump using PVSYST	https://www.youtube.com/watch?v=guL8Ussc5Cg	
Module 10 - Site Feasibility Study of solar PV water pump	Unit 10.2	Solar Pumping: A Cheaper and Cleaner Way to Access Groundwater	https://www.worldbank.org/en/news/video/2016/12/19/solar-pumping	
Module 11 - Installation of Solar PV water pump	Unit 11.1	solar water pumping	https://www.youtube.com/watch?v=x_vW6YniJjY	
Module 11 - Installation of Solar PV water pump	Unit 11.2	Shakti Solar Mounting Structure Installation Video	https://www.youtube.com/watch?v=DbJk1M4U7W4	
Module 11 - Installation of Solar PV water pump	Unit 11.2	Comprehensive Deep Well Rig Drilling Process	https://www.youtube.com/watch?v=lbsznSdhE9Y	
Module 12 - Operation and maintenance of Solar	Unit 12.1	Anti-Islanding Protection	https://www.youtube.com/watch?v=Ulot5MUJFbY	
Module 13 - Solar PV Water Pump Project lifecycle		Solar module BOM (Bill of materials)	https://www.youtube.com/watch?v=TraDKXKvPa8	

Module 14 - Site Feasibility Study for Ground Mount Solar power plant	Unit 14.1	What is Albedo & Why it is Important	https://www.youtube.com/watch?v=VhvwpxXnOf8	
Module 14 - Site Feasibility Study for Ground Mount Solar power plant	Unit 14.2	How to conduct SPT	https://www.youtube.com/watch?v=DjWDOqQjsyQ	
Module 15 - Interpret and prepare Civil foundation, layout, mounting structure and allied civil work in ground mount solar plant	Unit 15.1	Planning and Building a Solar PV Farm	https://www.youtube.com/watch?v=C2xYV57HvX8	
Module 15 - Interpret and prepare Civil foundation, layout, mounting structure and allied civil work in ground mount solar plant	Unit 15.2	CIVIL WORKS AND FOUNDATIONS FOR SOLAR PLANTS	https://www.youtube.com/watch?v=micjDafdy1w	
Module 16 - Planning and Scheduling	Unit 16.1	How a Solar Farm is Constructed from Beginning to End	https://www.youtube.com/watch?v=sM7ge9ahsGY	
Module 17 - Electrical Feasibility Study for Ground	Unit 17.1	Installing PV Wire And Conduit For Our Solar System	https://www.youtube.com/watch?v=kazwzLyKD2A	

Mount Solar Power Plant				
Module 17 - Electrical Feasibility Study for Ground Mount Solar Power Plant	Unit 17.1	Transformer information in solar plant	https://www.youtube.com/watch?v=oeJDOznZylc	
Module 18 - Manage Electrical installation in Ground Mount Solar Plant	Unit 18.1	Earthing Layout of electrical substation	https://www.youtube.com/watch?v=c2TXqWG2ld4	
Module 19 - Manage Electrical installation in Ground Mount Solar Plant	Unit 19.2	Application Process for CEIG Approval	https://www.youtube.com/watch?v=PIOwZWMSBVY	



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